

DELAWARE COUNTY AP, IN, USA

WMO: 725336

Lat: 40.234N

Lon: 85.394W

Elev: 937

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 94895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	6.7	-6.6	4.0	2.3	-1.5	5.2	7.6	26.1	30.2	24.2	30.0	7.5	270	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	90.4	73.5	88.2	72.9	85.9	71.6	77.1	86.2	75.5	83.8	74.3	82.4	8.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	131.3	82.7	72.8	125.8	81.1	71.8	121.7	79.8	41.1	86.0	39.6	84.4	38.3	82.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	20.1	18.1	DB	-4.7	93.4	6.4	3.7	-9.3	96.0	-13.0	98.2	-16.6	100.3	-21.3	103.0	(4)
			WB	-5.4	79.0	6.1	1.7	-9.8	80.2	-13.4	81.2	-16.9	82.2	-21.3	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	27.5	30.4	41.0	52.9	63.5	71.8	74.2	72.1	66.3	54.8	43.2	32.5
	DBStd	19.01	12.43	11.95	11.44	9.75	8.86	6.34	5.62	5.35	7.52	9.53	9.92	10.97
	HDD50	2534	702	559	330	79	6	0	0	0	1	56	251	550
	HDD65	5502	1163	970	748	378	139	17	3	7	73	343	654	1007
	CDD50	3494	4	9	50	167	426	655	750	685	489	204	47	8
	CDD65	988	0	0	3	15	94	222	288	227	112	26	1	0
	CDH74	8571	0	0	17	136	807	1855	2626	1873	1048	206	3	0
	CDH80	2653	0	0	1	15	223	605	893	533	334	49	0	0
Wind	WSAvg	8.5	10.5	10.3	10.0	10.2	8.5	7.4	6.2	5.6	6.1	8.0	9.3	9.8
Precipitation	PrecAvg	41.8	2.6	2.3	3.0	4.0	4.8	4.7	4.4	3.6	3.2	3.0	3.3	2.8
	PrecMax	56.7	7.7	5.3	6.3	6.9	7.5	9.2	11.3	8.4	8.1	7.0	7.5	8.0
	PrecMin	31.2	0.8	0.7	0.7	1.2	1.1	0.8	1.9	1.1	0.4	1.2	1.2	0.2
	PrecStd	6.3	1.4	1.2	1.4	1.6	1.6	2.0	2.2	1.7	1.8	1.5	1.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	65.6	76.3	82.0	89.5	92.2	93.8	90.5	91.5	85.5	72.4	63.2
		MCWB	55.8	54.2	61.8	64.6	69.6	72.6	77.0	74.6	72.3	69.7	60.2	56.8
	2%	DB	55.1	59.6	69.5	78.5	85.3	89.7	90.2	88.2	87.9	79.5	66.5	57.2
		MCWB	51.2	54.2	58.0	62.1	67.6	71.5	75.4	74.4	71.2	64.9	57.2	53.4
	5%	DB	50.5	53.9	63.8	73.9	82.2	86.5	87.8	85.5	83.9	74.8	63.0	52.6
		MCWB	47.0	49.3	54.6	60.0	66.6	70.8	74.3	72.7	69.7	63.2	55.7	49.0
	10%	DB	45.0	47.8	59.2	69.9	78.8	83.6	85.0	82.8	80.5	71.0	58.8	47.8
		MCWB	40.8	42.6	51.9	58.0	65.8	69.3	72.8	71.7	67.5	61.3	52.1	44.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	59.1	62.6	67.9	73.0	76.7	79.4	78.2	75.4	71.7	63.2	58.4
		MCDB	59.7	63.5	74.6	78.6	83.0	86.9	89.4	87.0	86.1	82.0	68.3	61.9
	2%	WB	52.1	54.4	59.5	64.7	70.7	74.8	77.7	76.4	73.5	68.9	59.8	53.6
		MCDB	54.5	57.9	66.7	73.9	81.1	84.4	87.0	84.8	83.0	76.5	64.9	56.4
	5%	WB	47.4	48.9	56.8	62.1	69.0	73.3	76.3	75.0	71.8	65.8	56.7	49.3
		MCDB	50.2	53.7	62.9	70.6	78.8	83.0	84.8	82.4	80.7	72.2	61.5	52.8
	10%	WB	40.9	42.6	52.7	59.6	67.4	71.8	74.8	73.4	69.8	62.2	52.8	44.4
		MCDB	44.8	47.3	58.3	68.0	76.1	80.2	82.7	80.2	76.6	69.3	58.7	47.3
Mean Daily Temperature Range	5% DB	MDBR	14.0	14.8	17.2	20.4	19.3	18.3	18.3	19.2	21.6	19.2	15.8	12.8
		MCDBR	17.9	20.7	22.3	24.6	22.7	21.0	19.3	20.3	24.8	23.3	20.2	16.9
		MCWBR	15.8	17.0	14.3	13.7	10.4	8.9	8.3	9.6	11.0	11.8	13.8	14.9
	5% WB	MCDBR	17.6	19.8	19.9	20.8	19.0	16.9	16.8	16.9	19.5	18.5	18.1	16.2
MCWBR		16.6	17.3	13.8	13.4	10.0	8.7	8.5	9.0	9.7	10.5	14.6	15.1	
Clear-Sky Solar Irradiance	taub	0.299	0.326	0.346	0.408	0.456	0.442	0.448	0.444	0.410	0.358	0.325	0.301	
	taud	2.416	2.324	2.359	2.192	2.100	2.195	2.183	2.182	2.252	2.396	2.441	2.444	
	Ebn at Noon	274	280	287	274	263	266	263	260	262	266	262	264	
	Edh at Noon	26	33	35	45	50	46	46	45	39	30	25	23	
All-Sky Solar Radiation	RadAvg	534	811	1142	1499	1724	1937	1923	1730	1413	978	644	444	
	RadStd	53	89	94	135	151	134	110	93	117	90	77	49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+340	+247
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	N/A

Nomenclature: See separate page